



SHEAT Group of Institutions
Learn Today, Lead Tomorrow

Branch - B.tech CS(AI\ ML)(B)

Topic - AI RAKSHA(Self
defence of girls through AI)

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I would express
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topic of self
defence of girls
through AI .
who also help me
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project

ABSTRACT

Women's safety is a major concern in india. In many emergency situations, girls are unable to call for help because they cannot operate their phones. The purpose of this project is to create smart safety using artificial intelligence for every category of girls poor or rich .

our system, AI RAKSHA, automatically detect danger and sends help. It uses voice recognition, shake detection, live gps tracking and AI based threat detection. When a girl is in danger and can't even take out her phone, just by shouting 'HELPS' or by shaking the phone, the system automatically sends live location to their trusted contact and police. It also includes fake incoming calls and alert for the dangerous areas and unsafe alert beeping start

INTRODUCTION

Every year ,thousands of cases are reported when girls do not help on time.Existing a SOS apps recquire to press button ,which is not possible in panic situation.

AI can solve this problem .AI understood human voice ,behaviour, and situation.our system is not just an app ,it is intelligent companion that monitrate 24*7.

Objective of this project:-

- 1.to provide instant help without phone
- 2.to enable real time location traking
- 3.to prevent incident before they happen through safe root guidance
- 4.to create low cast system that run on every's girls smartphone

PROBLEM STATEMENT & NEED for the SYSTEM

LIMITATION OF EXISTING SYSTEM

- * User has to manually press a panic button.
- * help from the police reaching late
- * Location sharing is manual and slow
- * No information about which area is safe

NEED FOR PROPER SYSTEM::

we need a system that can understand without touch, only by voice and movement, that a girl is in danger automatically call for help. This will save crucial crime and save lives.

LITERATURE SURVEY

We studied 3 existing and research paper ::

- 1.112 India app's : government's official emergency app, but SOS has to be triggered manually.**
- 2. bsafe, shake2safety : they have a shake feature but do not use AI, so they contain many false.**
- 3. IEEE Research paper(2023) - Women safety using ML :: In this paper, YOLO model was used for harassment detection through CCTV.**

By removing the drawbacks of these system, we have created a combined system of AI voice trigger + IoT+GPS

PROPOSED SYSTEM ::

Our system is build with 4 modules ::

1. Voice trigger engine :
when the user says 'HELP',' BACHAO' 'save me' ,the whisper Ai. mode detects it with 98% accuracy , even in crowed area.
2. Shake and Fall detection ::
Uses the accecleromotor sensor of the smartphone.shaking the 3times quickly will triger the SOS .
3. Live tracking and Alert system::
 - * SMS + Whatsaap alert to 3 trusted contact via Twillio API.
 - * live location link with google Maps.
 - * 10 sec audio recording of the surrounding
 - * Auto email to the nearest police station.
3. Safe Route AI ::
suggest a safe route at night using google maps API + crime data.if users fell dangerous area then ,it give vibrates loudly alert

SYSTEM ARCHITECTURE & FLOW CHART

ARCHITECTURE::

User layer(android app)->AI
processing layer (speech model
+ logic layer)->cloud layer
(firebase - Real time Database
for location)->Alert layer
(Twilio,Whatsaap API ,Email service).

Flowchart::

START ->App is ON in
background->waiting for voice /
shake input->Is threat detected?->if
NO,continue waiting->if YES, capture
location + audio -> send alerts to
contact and police -> helps arrives
and STOP

TECHNOLOGY STACK AND METHODOLOGY ::

-

Frontend : flutter /android studio(java/
Kotlin)
Backend : Firebase Realtime
Database,Firestore.
APIs used : google maps APIs ,Twilio
SMS and Whatsaap API

AI MODELS:

- * speech recognition : open Ai
whisper/google search -to-test
- * keyword sppoting : Tensorflow lite
custom model
- * Threat Detection(optional for cctv
version): YOLOv8

Hardware : panic button
Meathodology : Agile button

CONCLUSION , FUTURE SCOPE AND REFERENCE

AI RAKSHA , is a low cost ,high speed safety solution that can be installed on every girls phone.it not only sends but also helps to tries to prevent the incident by guiding through safe routes.

Future scope ::

1. Integrate the system in smartphone /bracelets/ smartwatch.
2. Direct connect the police control room or personal contact .
3. AI- based face recognition of stalkers/repeat offenders
4. Drone support in live tracking in future.

THANK YOU 😊

AI can put women
and girls at risk —
or help protect
their rights.





HER OWN HERO

THE ORIGINS OF THE WOMEN'S
SELF-DEFENSE MOVEMENT



WENDY L ROUSE

Crimes against women: 90% of trials pending in city, says Pragna
 Mumbai Report

Crime against women, kids: Conviction lowest in 3 years

Mumbai Report

By [Name]

RAPE CASES UP 24% IN CITY OVER 5 YRS

Crimes against women and children have increased in Mumbai over the last five years. According to a report by the Mumbai Police, the number of rape cases has increased by 24% over the last five years. The report also states that the conviction rate for these cases is the lowest in three years.

Mumbai News

CRIME IN MUMBAI, BY REGION

(Data for 2018-19, 2019-20, 2020-21)



RAPE CASES

Investigation



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'Awareness camps saved 90 trafficked girls last month'

Dwaipayan Ghosh
@timesofindia.com

N BENGAL, SUNDARBANS AT HIGH RISK

Kolkata: Bengal reported approximately 67 cases of trafficking according to the last NCRR report tabled in 2022. The report made it clear that migrants from vulnerable regions often become targets for traffickers, who lure them with false promises of employment and push them into exploitative conditions.

Multiple awareness camps at the ground level by the state, NGOs, and central agencies from schools to railway stations have helped save more than 90 women and girls in the past month alone. The RPF Just Rights for Children (with 400 NGOs under one umbrella) combination has just begun tasting success here, just like in a few other parts of India," said a spokesperson for Shakti Vahini, an NGO that works with trafficking victims across Bengal.

According to NGO workers, the top concern areas remain north Bengal and the Sundarbans. To try and negate illegal trafficking, multiple efforts have come together with state and central

Bengal reported 67 trafficking cases in the 2022 NCRR report

Migrants from vulnerable regions are often targeted by traffickers with false job promises

North Bengal and Sundarbans are identified as high-risk areas for trafficking

NGOs have partnered with state and central agencies



in Alipurduar, Jalpaiguri, Kalimpong, and Barapeta police district to combat trafficking

Awareness camps in schools and railway stations have helped rescue over 90 women and girls in the past month

Raids conducted at bus stands and NTP station have been part of successful operations

The RPF-Just Rights for Children initiative, involving 400 NGOs, has been effective in tackling trafficking

The joint efforts by state police, along with RPF and SSB, are also using technology, like facial recognition systems at railway stations, to recognise habitual sexual offenders. This is how a girl trafficked from Bengal to the Andamans was rescued in 24 hours, said cops. A local NGO alerted police, who contacted the NGO at the destination, and everyone acted.

Several local efforts, police and NGO sources said, have also worked in the past year

the most important to "break the chain", they felt. "Every trafficked child is part of a chain: source, transit, and destination. The trafficker is part of a system. The only way to stop the system is to break every link. That means prosecuting recruiters, transporters, and buyers, not just employers. We must cut off trafficking by making economically unviable trafficking properties, cancelling lines, cancellation, blacklisting